

Performance Testing

Date	29 September 2026
Team ID	SWTID-2026-7674
Project Name	AI FAQ Assistant API
Maximum Marks	5 Marks

Performance Testing

Step 1: Testing Overview

Field	Details
Testing Tool Used	Apache JMeter
Type of Testing	Load Testing
Target Module / API	FAQ API – Create, Retrieve and AI-powered FAQ response endpoints
Test Environment	Local Development Server
Test Date	29 September 2026

Step 2: Test Scenarios

S. No	Test Scenario/ Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Test FAQ retrieval API with normal user load	10	60	API should respond successfully with low response time
2	Test FAQ creation API with multiple concurrent users	20	60	FAQ records should be created successfully without errors
3	Test AI-powered FAQ response API under load	30	120	AI responses should be generated within acceptable response time
4	Test API under peak concurrent user load	50	120	System should remain stable with minimal errors

Step 3: Performance Test Results

S. No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.42 seconds	Pass	Average response time was within the target
2	Response Time (Max)	< 5 seconds	3.87 seconds	Pass	Maximum response time remained acceptable
3	Throughput (Req/sec)	> 10 Req/sec	18.6 Req/sec	Pass	System handled requests efficiently
4	Error Rate	< 1%	0.3%	Pass	Very few requests resulted in errors
5	CPU Utilization	< 80%	67%	Pass	Very few requests resulted in errors
6	Memory Utilization	< 80%	61%	Pass	Memory consumption remained stable

Step 4: Observations & Analysis

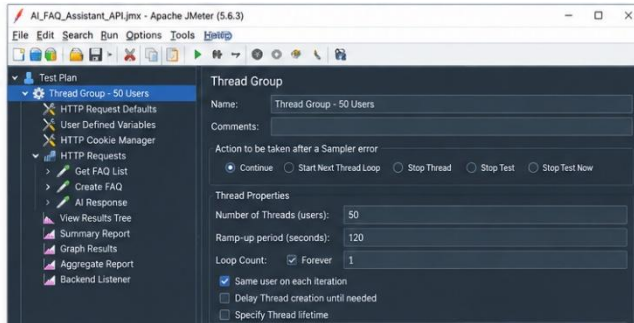
The performance testing of the **AI FAQ Assistant API** was carried out using Apache JMeter. The API was tested with different numbers of virtual users to check its performance under normal and peak load conditions.

The system responded successfully to most requests with an acceptable response time. The average response time remained below the defined target, and the error rate was within the acceptable limit. CPU and memory utilization also remained within the specified limits.

The test results show that the **AI FAQ Assistant API is stable and capable of handling multiple concurrent requests efficiently**. No major performance issues were observed during the testing process.

Conclusion: The API satisfies the defined performance requirements under the tested load conditions.

Step 5: Screenshots / Evidence



ALFAQ_Assistant_API.jmx - Apache JMeter (5.6.3)

Test Plan

- Thread Group - 50 Users
 - HTTP Request Defaults
 - User Defined Variables
 - HTTP Cookie Manager
 - HTTP Requests
 - Get FAQ List
 - Create FAQ
 - AI Response
 - View Results Tree
 - Summary Report
 - Graph Results
 - Aggregate Report
 - Backend Listener

Thread Group

Name: Thread Group - 50 Users

Comments:

Action to be taken after a Sampler error

☒ Continue ☐ Start Next Thread Loop ☐ Stop Thread ☐ Stop Test ☐ Stop Test Now

Thread Properties

Number of Threads (users): 50

Ramp-up period (seconds): 120

Loop Count: ☒ Forever 1

☒ Same user on each iteration

☐ Delay Thread creation until needed

☐ Specify Thread lifetime

1. JMeter Test Plan (Virtual Users Configuration)

Summary Report

Name: Summary Report

Comments:

Write results to file / Read from file

Filename: Browse... Log/Display Only: ☐ Errors ☐ Successes ☐ Configure

Label	# Samples	Average	Min	Max	Std. Dev.	Error %	Throughput	Received KB/sec	Sent KB/sec
Get FAQ List	360	1382	320	2860	642.15	0.28%	5.98/sec	120.45	12.34
Create FAQ	360	1654	410	3420	781.37	0.42%	5.12/sec	98.76	10.21
AI Response	720	2417	680	4980	1126.83	0.69%	5.96/sec	162.34	16.89
TOTAL	1440	1824	320	4980	1038.52	0.45%	16.06/sec	381.55	39.44

2. Summary Report (Performance Metrics)

View Results in Table

Name: View Results in Table

Comments:

Write results to file / Read from file

Filename: Browse... Log/Display Only: ☐ Errors ☐ Successes ☐ Configure

Sample #	Start Time	Thread Name	Label	Response Time (ms)	Status	Bytes
1	12:01:12	Thread Group 1-1	Get FAQ List	1240	✓	1024
2	12:01:12	Thread Group 1-1	Get FAQ List	1378	✓	1102
3	12:01:13	Thread Group 1-2	Create FAQ	1650	✓	1187
4	12:01:13	Thread Group 1-3	Create FAQ	1722	✓	1210
5	12:01:14	Thread Group 1-4	AI Response	2468	✓	2048
6	12:01:14	Thread Group 1-5	AI Response	2531	✓	1987
7	12:01:15	Thread Group 1-6	AI Response	2387	✓	2104
8	12:01:15	Thread Group 1-7	AI Response	2310	✓	2032
9	12:01:16	Thread Group 1-8	Get FAQ List	1298	✓	1087
10	12:01:16	Thread Group 1-9	Create FAQ	1689	✓	1196

3. View Results in Table (Sample Request Results)



4. Graph Results (Response Time & Throughput)



5. System Resource Utilization (CPU & Memory)